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Unit 3

How to write RESULT

Teaching Team of Ethics and Scientific Writing Technique Course

Faculty of Engineering
Universitas Gadjah Mada, 2025

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Contents

1. **Common model**
2. **Present your results**
3. **Build a model of the Result section**
   - ✓ Using target articles to build a model
   - ✓ Example of the excercise



https://lantsandlaminins.com/writing-guides/writing-guide-for-introductions-to-manuscripts-or-project-reports/

The Shape of a Research Article

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TODAY OFFER

Prepare Figures/Graphs

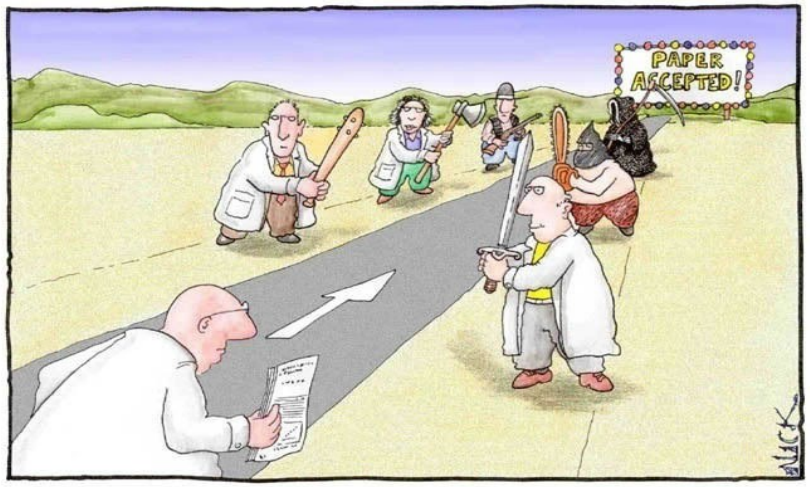
Prepare Results/Discussion

Prepare Methodology ✓

Prepare Introduction ✓

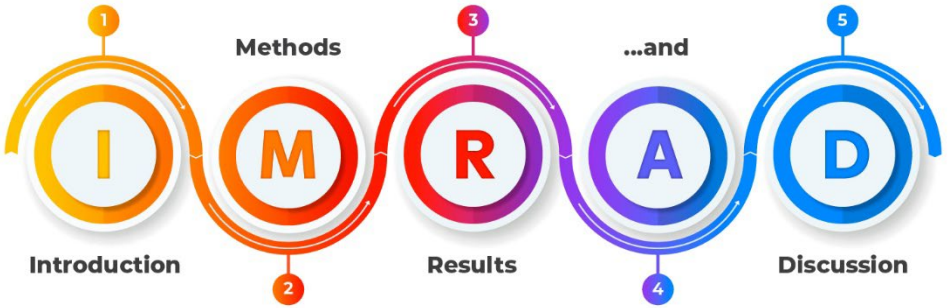
Prepare Abstract & Title

Order of our work when we prepare our manuscript



Most scientists regarded the new streamlined peer-review process as 'quite an improvement.'

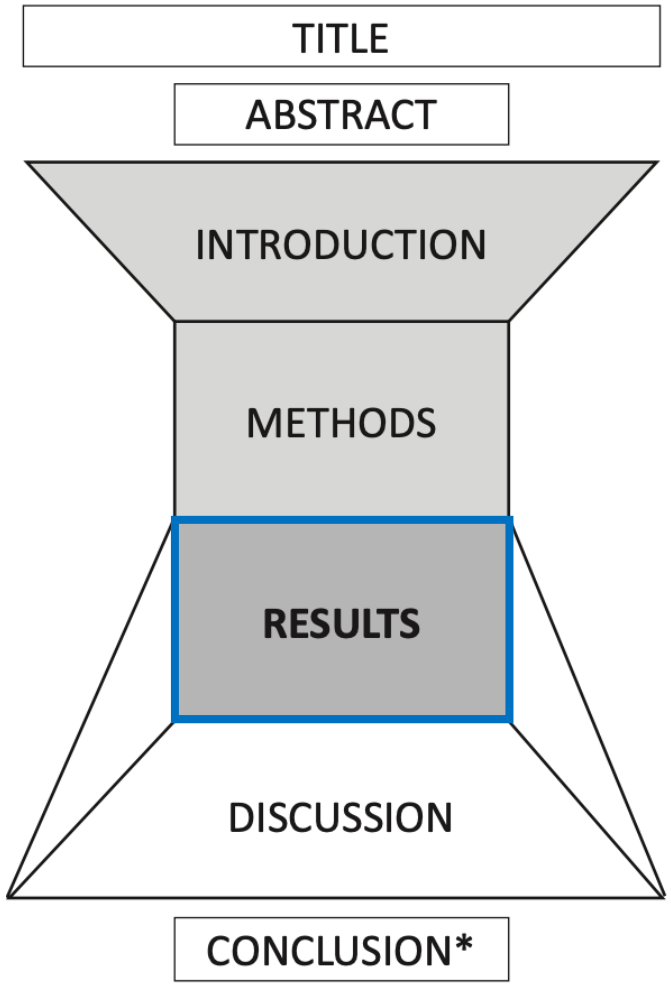
www.weirdscience.ca



1

Common Model

The structure and content of the results



Four common options for the final sections of the research article

SUBTITLES Option 1	SUBTITLES Option 2	SUBTITLES Option 3	SUBTITLES Option 4
Results	Results	Results and Discussion	Results
Discussion	Discussion		Ø
Conclusion	Ø	Conclusion	Conclusion

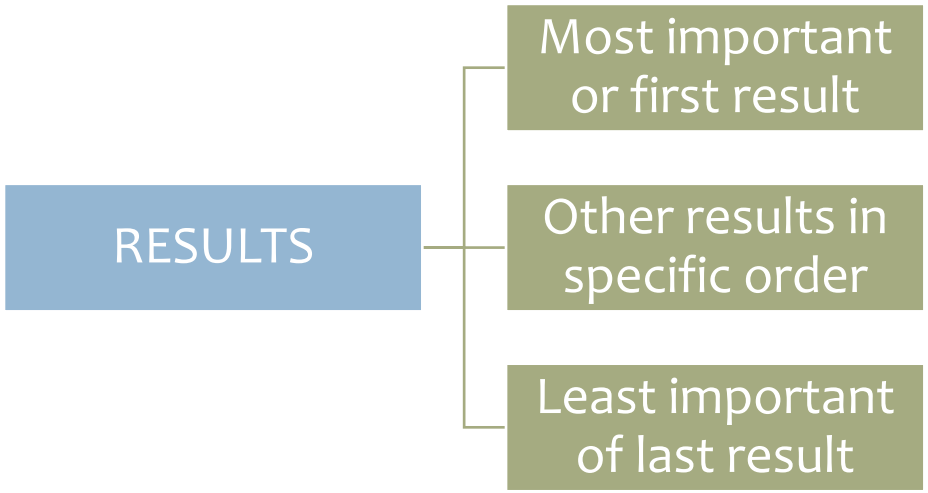
Choosing the Best Option

❑ **Check your target journal**
- Some journals have fixed structures.
- Others offer flexibility—follow examples in your target articles.

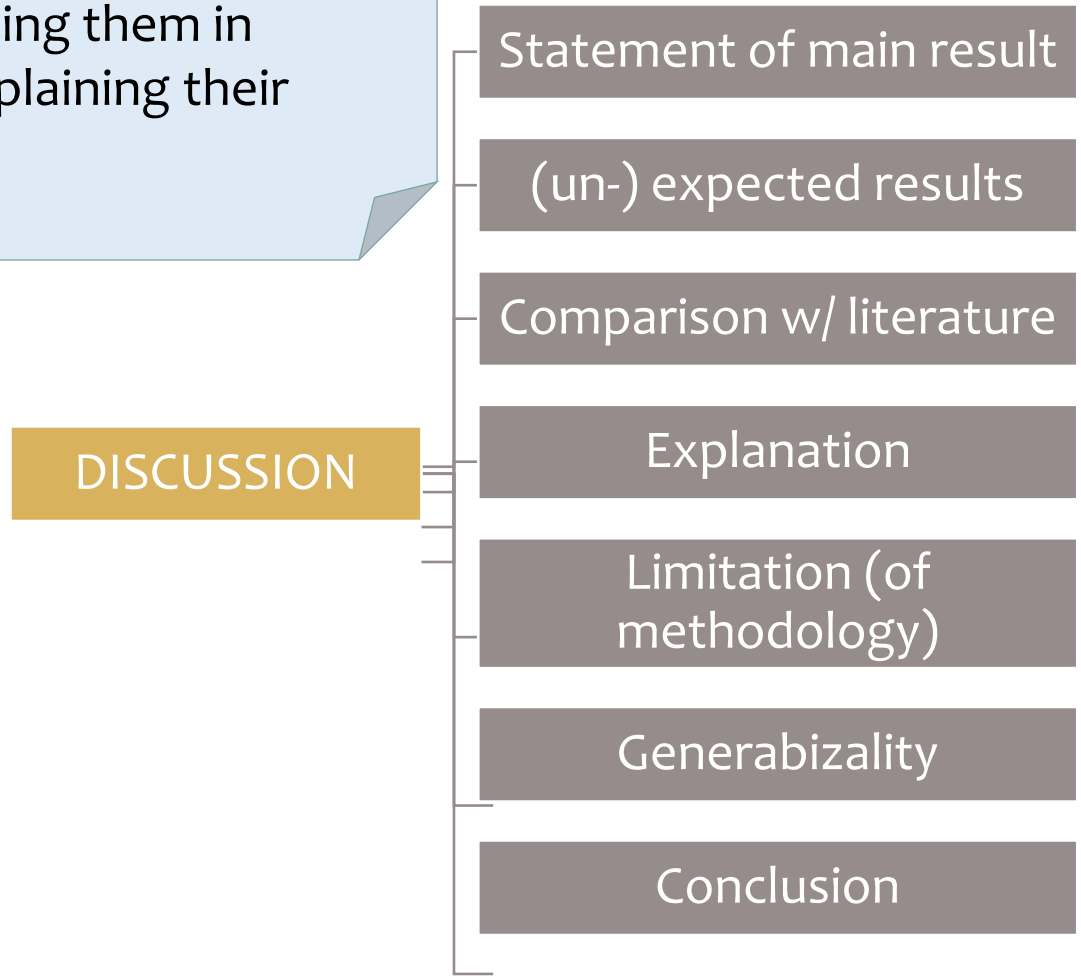
❑ **Let your study's nature guide the choice**:
- If **preliminary results influence later results** → combined *Results and Discussion* may work better.
- If results address **multiple hypotheses/predictions** → combine results & discussion for each hypothesis, with clear subtitles.
- If you want to **discuss results as a whole** → keep *Results* and *Discussion* separate.

What is the difference between Results and Discussion in Academic Writing?

The results section objectively reports your findings as they are; no speculations on why you found the results.



The discussion section interprets the results, putting them in context, and explaining their importance.



Results vs Discussion

Results section	Discussion section
This section answers the question “What?” about your research	This section answers the question “So what?” about your research
Describes the experiments conducted before writing the paper	Summarizes and interprets the significance of the key findings
States the results but does not interpret them	Interprets the results but does not re-state them
Includes only data that will be important to the Discussion section	Does not present new results, so do not make statements that your results cannot support
Uses the past simple	Uses both past and present tense as needed
It may include non-textual elements such as tables, pictures, and photographs	Use only text, although you may also refer to non-text elements

https://science-publisher.org/how-to-write-the-results-and-discussion-sections-in-a-research-paper/

2

Present your results

Present your results

A logically organized and written Results section is critical for a well-written paper. The Results section should present the data collected during the study as objectively, logically, and concisely as possible. Highlighting the most important results or organizing them into sections shows that you have considered all the necessary information. Proper use of visual elements, i.e., tables and figures, engages your readers.

So, essentially, in the Results section, you should do the following:

- State only the actual results: leave the explanations and comments for the Discussion section;
- Use the text, tables, and figures to highlight key findings in an orderly fashion;
- Ensure that the contents of the tables and figures do not repeat in the text.

https://science-publisher.org/how-to-write-the-results-and-discussion-sections-in-a-research-paper/

Choosing the proper format for presenting the results

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Use the table	Use the figures	Use the text
If you want to publish many exact numeric values and other data so that they take up little space	Figures are best for presenting trends, samples, and relationships between data sets	If the data reported are not too big or complex
If you want to compare and contrast the data values or characteristics of items related to each other or with several general features or variables	Graphs, pie charts, and data maps are good choices if you want to summarize the results of your research	If you need only 1 or 2 columns to present the data in a table
If you want to show the presence or absence of specific characteristics	To visually present a sequence of phenomena, characteristics, events, or geographic indicators	When reporting data that are irrelevant to your main results but still has some significance

https://science-publisher.org/how-to-write-the-results-and-discussion-sections-in-a-research-paper/

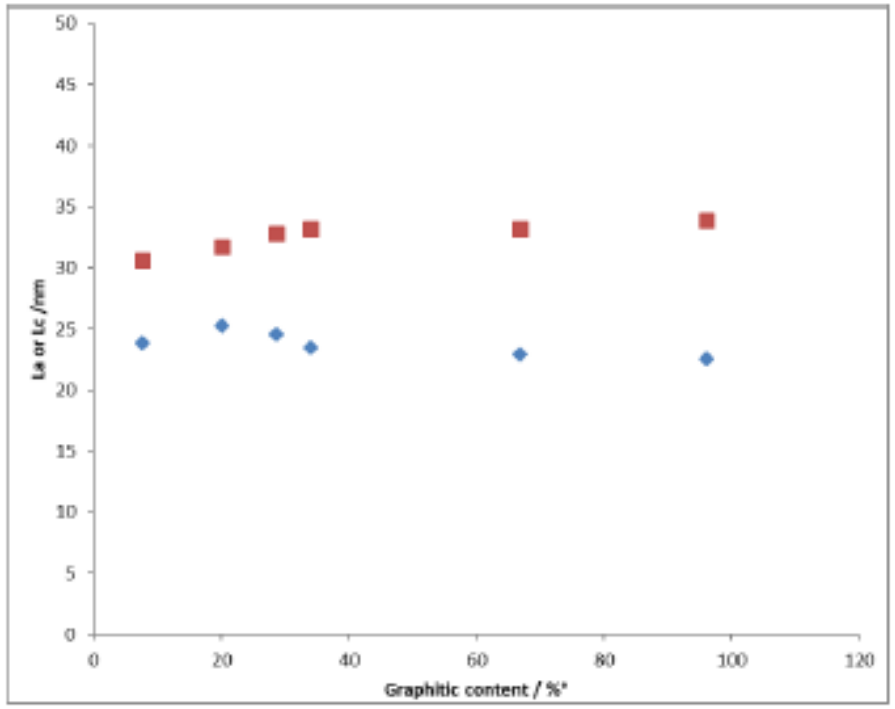
Prepare your figures

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- ❑ Provide clear and high quality figures !
- ❑ Preferred format : EPS, EMF, TIFF, JPEG with the format of 300 DPI. Do not provide figures with a low-resolution format like GIF, BMP and WPG.
- ❑ Please be serious to prepare your graphs ! Allocate your time sufficiently.
- ❑ There are many standard softwares to help you prepare your figures. See the list:

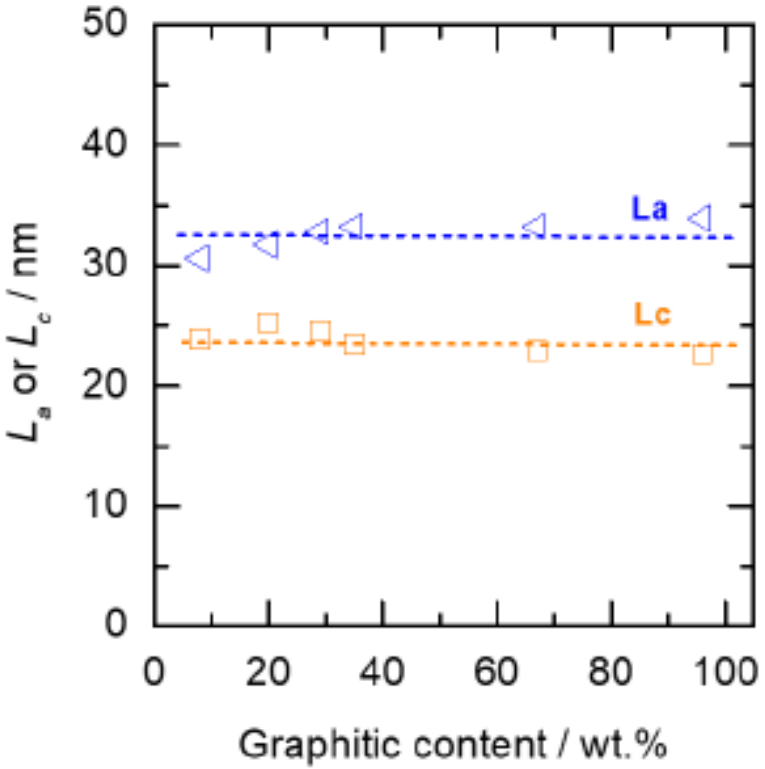
  https://en.wikipedia.org/wiki/List_of_information_graphics_software

Prepare your figures



bad figure!

- The font size is too small
- Minimum effort → low resolution and minimum information
- Low resolution



A good one

Feel free to comment why this is a good figure…..

...Choosing the proper format for presenting the results

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Benazir & Oktari (2024)

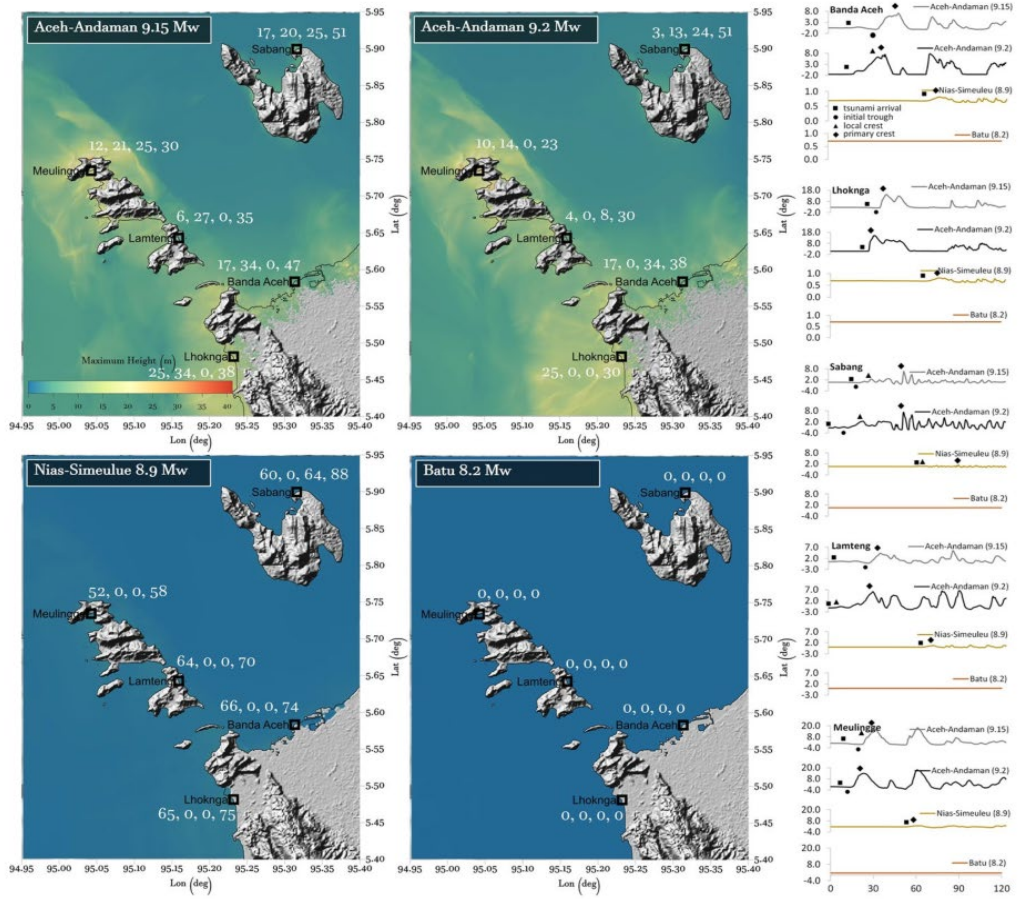


Fig. 7 Characteristics of tsunami arrival phases for Northern Aceh coast (tsunami arrival, initial trough, local crest, primary crest: minute)

Syamsidik, Benazir et al. (2020)

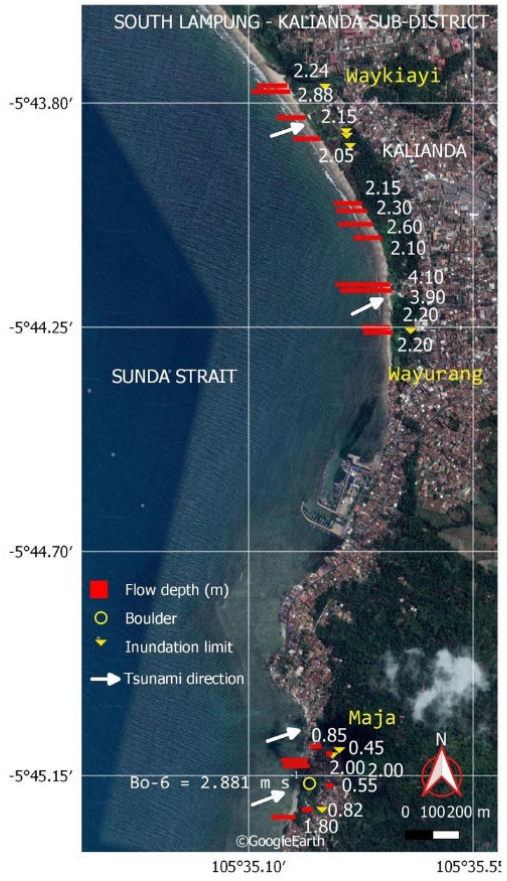


Figure 9. Flow depth distribution and run-up limits in Kalianda, South Lampung. Green bars and yellow triangles represent flow heights and inundation limits measured at the locations, respectively. (The base map was adapted from Google Earth.)

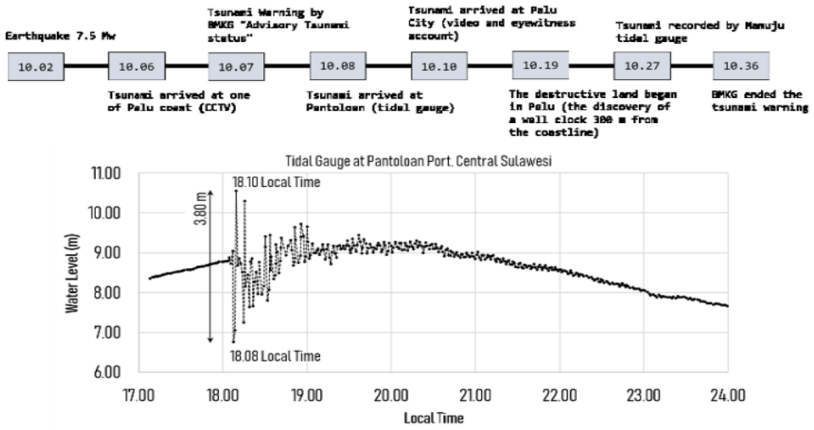


Fig. 1 The timeline event and the recorded tsunami fluctuation by a tidal gauge at Pantoloan Port (BIG, 2018)

Benazir et al. (2020)

...Choosing the proper format for presenting the results

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Table 2. *Vegetation characteristics in Pacitan Bay based on field inventory results.*

No.	Sector*	Area (m^2)	$Di_{average}$ (m)	$S_{average}$ (m)		Open gap (m)	Species
				x	y		
1	A	3,653	0.29	4.20	3.76		*C. equisetifolia*
						42.78	
2	B	3,709	0.25	3.92	4.16		*C. equisetifolia*
						323.53	
3	C	11,054	0.17	3.06	2.91		*C. equisetifolia*
						10.74	
4	D	8,619	0.09	4.78	4.87		*C. equisetifolia* and *P. odorifer*
						12.42	
5	E	37,085	0.18	3.44	3.19		*C. equisetifolia* and *P. odorifer*
						102.45	
6	F	37,258	0.19	2.28	2.33		*C. equisetifolia* and *P. odorifer*
						12.23	
7	G	78,006	0.24	2.15	1.91		*C. equisetifolia* and *P. odorifer*
						33.96	
8	H	18,664	0.17	2.00	2.33		*C. equisetifolia*
						6.83	
9	I	34,843	0.25	3.94	3.13		*C. equisetifolia* and *A. auriculiformis*
						23.40	
10	J	1,462	0.02	1.20	1.20		*C. equisetifolia*

$Di_{average}$ = average of tree diameter (m), $S_{average}$ = average of tree distance (m), *refers to figure 2.

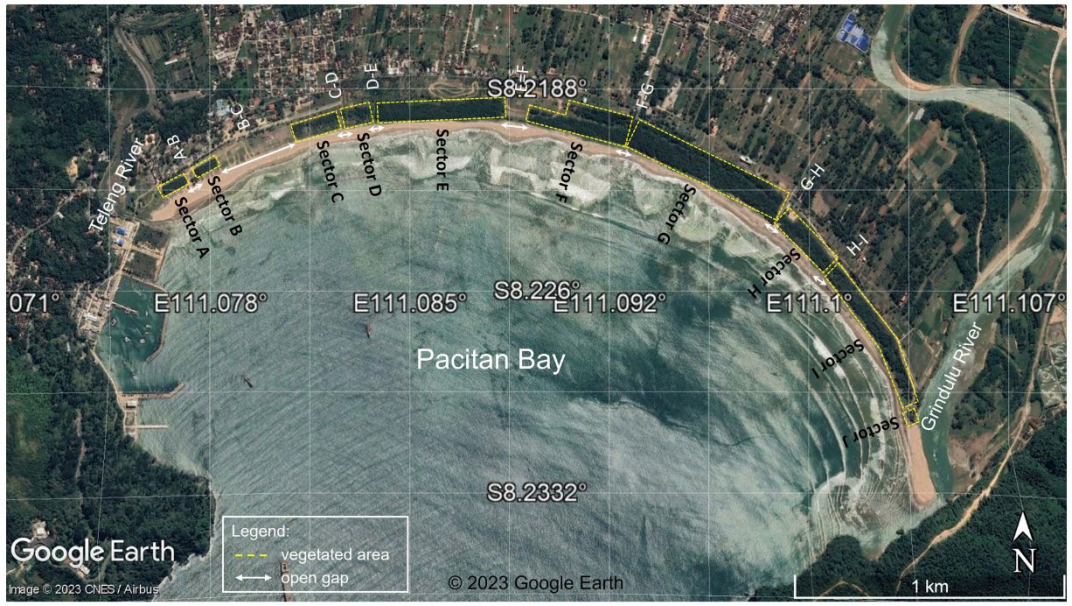


Figure 2. The sectors and open gaps of vegetated area in Pacitan Bay.

Benazir et al. (2024)

3

Building a Model (based on Glasman-Deal 2021)

Building a model to write Results

Key Message

- Don’t be rigid with outdated rules
- Observe and learn from current articles
- Develop your own writing model for Results
- Keep updating and adapting your competence over time



Using Target Articles to Build a Model

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Why Write a Results Section?

- Research results can be shown in:
  - Graphs, tables, equations, photographs, images
- Question: Why not stop there?
- Answer: Graphics alone are **not enough**
  - Readers need **narrative text** to:
    - Explain results
    - Highlight key points
    - Guide interpretation



Example for today exercise

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On-bicycle exposure to particulate pollution: using a portable device for accurate measurement

Results

1 *Data obtained in previous cycling safety studies[1,5,7] used fixed monitors positioned at high-emission locations to measure weekday concentrations of black carbon (BC) and nitrogen dioxide (NO_2).* ***2*** *Those studies found no significant difference in BC and NO_2 emissions between cycle lanes shared with buses and cycle lanes separated from vehicle traffic.* ***3*** *In our study, BC and NO_2 concentrations were measured using a PEMS-43, a portable emissions measurement system developed by Noxious et al.[7].* ***4*** *The PEMS-43 was fixed to a bicycle and recorded the emissions hourly along each route type.*

5 *Figure 1 shows the BC and NO_2 concentrations measured hourly on each route type from 6.00 a.m. to 10.00 a.m.* ***6*** *In line with results from previous studies, the BC and NO_2 concentrations were broadly similar for both route types over this 4-hour period.* ***7*** *However, the data obtained by PEMS-43 for journeys later in the day were significantly different.* ***8*** *We recorded a striking reduction in the NO_2 concentrations along bike paths separated from vehicle traffic, beginning shortly after 10.00 a.m. (Figure 2).* ***9*** *As can be seen, by 2.00 p.m. the NO_2 levels had fallen by as much as 33%, and this low level was maintained even during the 5.00 p.m. to 7.00 p.m. peak traffic density period.* ***10*** *These results suggest that NO_2 levels along bike paths separated from vehicle traffic may not match those of BC at all times throughout the day.*

11 *The presence of vegetation is known to slightly alter BC and NO_2 concentrations at different times of day[11,18], and the effect of this along the two routes was not investigated.* ***12*** *Nevertheless, the data obtained here suggest that using a portable emissions measurement system to measure BC and NO_2 exposures hourly may provide more accurate information for traffic management strategies than traditional on-site measurement.*

How to begin the Results section?

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In Sentences 1 and 2 *Data obtained in previous cycling safety studies[1,5,7] used fixed monitors positioned at high-emission locations to measure weekday concentrations of black carbon (BC) and nitrogen dioxide (NO_2). Those studies found no significant difference in BC and NO_2 emissions between cycle lanes shared with buses and cycle lanes separated from vehicle traffic.* **the writer repeats/revisits the methods and findings of studies mentioned earlier in the article.**

In Sentences 3 and 4 *In our study, BC and NO_2 concentrations were measured using a PEMS-43, a portable emissions measurement system developed by Noxious et al.[7]. The PEMS-43 was fixed to a bicycle and recorded the emissions hourly along each route type.* **the writer briefly summarises the method used in this study.**

Why Start Results by Revisiting Earlier Sections?

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⬇ Smooth Transition
Connects previous section to the new section.

⬇ Same Starting Point
Reminds readers of the gap/problem & key studies.

⬇ Help Skimmers
Some jump straight to Results from Title/Abstract.

⬇ Recall Key Methods
Important if Methods are long or in supplement.

⬇ You Know It, They Don't
Readers may skip or forget the Methods.

Why Not Just Present Results in Order?

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⬇ Not a Project Timeline

A paper's goal is contribution & clarity — not chronological storytelling.

⬇ Readers Seek Patterns

If listed in order, readers must keep re-adjusting their view of trends.

⬇ You Know the Path, They Don't

Without guidance, your direction may be unclear.

⬇ Risk of Misinterpretation

Different readers may see different patterns — possibly misleading.

⬇ Your Role as Guide

Organise results in a narrative that leads toward your conclusions.

Good Ways to Begin the Results Section

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⬇ Check Your Target Journal

See how successful articles in your field start their Results.

⬇ Start with a General Statement

Summarise overall trends/patterns before details. → Helps readers see the “wall” before the “bricks.”

⬇ Refer to a Key Figure/Table

Guide readers to a summary visual early on.

⬇ Re-state the Aim of the Study

Sets clear context for evaluating results. → Helps you decide what to include and in what order.

⬇ Recall Key Aspects of Method

Especially useful if Methods are long or in Supplementary Materials. → Supports readers who skip directly to Results.

Why Invite the Reader to Look at the Graphics?

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In Sentence 5 *Figure 1 shows the BC and NO_2 concentrations measured hourly on each route type from 6.00 a.m. to 10.00 a.m.* **the writer invites the reader to look at a Results graphic.**

Why Invite the Reader to Look at the Graphics?

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⬇ Readers Look Anyway
They often pause reading to study graphics before continuing.

⬇ Guide Their Interpretation
Frame graphics so readers see them in the intended context.

⬇ Avoid Random Placement
Graphics need to appear in a logical, purposeful sequence.

⬇ Plan Narrative + Visual Order
Think about why each graphic is placed where it is, and how it leads to key results.

⬇ Build Towards the Big Results
Sometimes it's more effective to start with supporting visuals, then reveal the important ones.

⬇ Prevent Misinterpretation
If visuals can be read in multiple ways, comment first before inviting the reader to look.

Do I Need to Compare My Results with Others?

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In Sentences 6 and 7 *In line with results from previous studies, the BC and NO_2 concentrations were broadly similar for both route types over this 4-hour period. However, the data obtained by PEMS-43 for journeys later in the day were significantly different*. **the writer compares the results with those in other studies, using subjective, evaluative language** *(In line with, broadly similar, significantly different)*.

…Do I Need to Compare My Results with Others?

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⬇ Crossing the Line to Original Research

Now writing for readers who don't know your exact methods, expected results, or comparisons.

⬇ Research = New Contribution

You must identify how your work advances the field.

⬇ Position Your Study on the Map

Comparisons help readers see where your study fits in the literature.

⬇ Compare in the Results Section

Show how your findings align with or differ from previous studies before discussing them in depth.

⬇ Types of Comparisons; With published literature: With your own predictions: Modelled vs. experimental results

⬇ Careful with Language

How you describe similarities/differences can affect how your work is understood.

Why Use Evaluative Language in Results?

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⬇ Results Don’t Speak for Themselves
Your text should guide readers toward the conclusions you’ll draw.

⬇ Shape Reader Perception
The words you choose influence whether readers focus on similarities, differences, strengths, or weaknesses.

⬇ Align Aim → Results → Conclusions
Without commentary, readers may interpret the data differently from you.

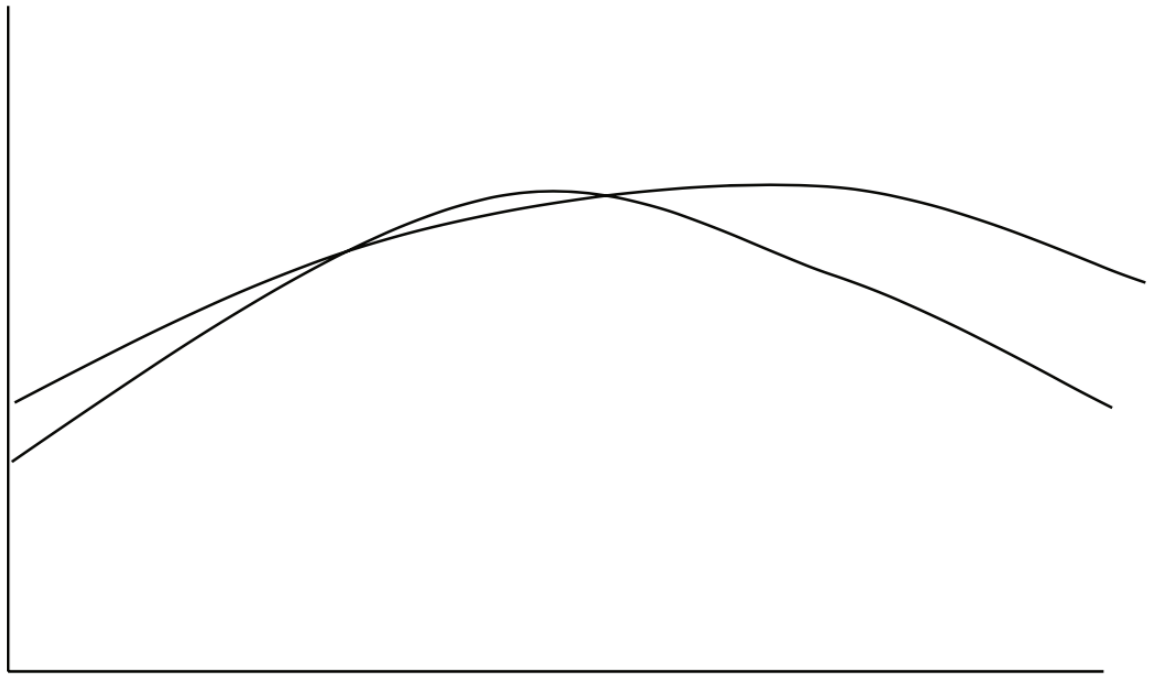
⬇ Avoid the “So What?” Problem
Simply stating numbers adds nothing beyond what’s visible in the graphic.

⬇ You’re Part of the Research Community
You have the right — and responsibility — to interpret your results.

⬇ Use Language to Signal Strength
The same number can be presented as strong (“in as many as 23%”) or weak (“in only 23%”).

⬇ Learn from Target Articles
Identify what evaluative terms are used, when, and how often.

…Why Use Evaluative Language in Results?



If you write “As can be seen in Fig. 3.1, the two curves are very similar,” readers focus on similarity;

if you write “… are significantly different,” they focus on difference.

Words/Phrases to Comment about Quantities

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Group 1: *words/phrases which 'increase' the size/quantity*

a great deal (of)	high	over (half/25%)
a number (of)	large	plenty of
abundant	marked	quite a few
appreciable	more (than)	quite a lot
as many as	most	several
at least	much	significant
considerable	noticeable	substantial
greater (than)	numerous	upwards of

Group 2: *words/phrases which 'reduce' the size/quantity*

a few	hardly	minor
a little	just	modest
as few as	less	only
barely	little	scarcely
below	low	slight
few	marginal	small
fewer (than)	minimal	under

Group 3: *words/phrases which emphasise how big/small the size/quantity is*

appreciably	far (more/less)	significantly
considerably	markedly	so (high/low)
easily (over/under)	much (higher/lower)	substantially
even (higher/lower)	noticeably	very
exceptionally	particularly	well (over/under)
extremely	remarkably	

Group 4: *words/phrases which communicate that the size/quantity is similar or close to another*

approximately	just (over/under)	practically
close (to)	little (i.e. close to none)	roughly
few (i.e. close to none)	minimal	slightly (more/less)
imperceptible	near/ly	virtually
	negligible	

Group 5: *words/phrases which you can use when you do not want to interpret the size/quantity*

fairly	quite (high/low)	some
in some cases	rather (high/low)	somewhat
moderate	reasonably	to some extent
	relatively	

Is ‘Striking’ Too Informal?

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In Sentence 8 *We recorded a striking reduction in the NO_2 concentrations along bike paths separated from vehicle traffic, beginning shortly after 10.00 a.m. (Figure 2).* **the writer directs the reader’s attention to a specific result, describing it with strongly evaluative language** *(striking).*

…Is ‘Striking’ Too Informal?

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⬇ Apparently Not
Science writers often use strong words instead of punctuation for emphasis.

⬇ No Exclamation Marks
Even for exciting results, formal writing avoids “!” marks.

⬇ Use ‘!-Substitutes’
Words like *striking* create a “wow” effect without breaking formal tone.

⬇ Learn the Vocabulary
Check your field’s common impact words and apply them appropriately.

How to Decide Which Results to Present in Detail

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In Sentence 9 *As can be seen, by 2.00 p.m. the NO_2 levels had fallen by as much as 33%, and this low level was maintained even during the 5.00 p.m. to 7.00 p.m. peak traffic density period*. **the writer selects a specific result to present in more detail, and comments on it** *(as much as, even)*.

…How to Decide Which Results to Present in Detail

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⬇ Guide, Don’t Just Describe
If you present all results equally, readers can’t see which ones matter most.

⬇ Not All Results Are Equal
Some are key findings, others are minor. Even key results vary in importance.

⬇ No Voice, No Red Ink, No “!”
Your writing must signal importance since emphasis tools are limited.

⬇ Highlight Key Results
Use phrases like *It is important to note that… / Importantly, / It is significant that…*

⬇ Best Results Framed as Typical
Writers often present strong findings as if they represent the overall pattern.

⬇ Example
The SFS results are generally in very good agreement with their FE counterparts; for example, at midspan the values are almost identical.

Should I Explain the Results as well as Comment on Them?

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⬇ Check Target Articles

Reverse-engineer how writers in your field handle explanations.

⬇ Explanations Are Common

Often included in Results sections, but usually concise.

⬇ Two Types of Explanation

How result was obtained → method details

Why result occurred → material/system properties

⬇ Distinguish Explanation vs. Implication

Explanation = direct account of the data

Implication = what data suggest → usually saved for Discussion

⬇ Balance Carefully

Brief implications may appear in Results, but full interpretation belongs in Discussion.

Do I Need to Mention Limitations or Problems?

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In Sentence 10 *These results suggest that NO_2 levels along bike paths separated from vehicle traffic may not match those of BC at all times throughout the day.* **the writer mentions a possible implication of the results.**

In Sentence 11 *The presence of vegetation is known to slightly alter BC and NO_2 concentrations at different times of day[11,18], and the effect of this along the two routes was not investigated.* **the writer mentions a possible limitation, minimising its potential impact** *(slightly)* **on the results.**

…Do I Need to Mention Limitations or Problems?

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⬇ Opposite of What You Fear

Acknowledging problems does not weaken your paper — it strengthens credibility.

⬇ Don’t Ignore Issues

If results are incomplete or don’t fit, mention them (unless they are truly insignificant and invisible).

⬇ Show You’re in Control

Recognising limitations demonstrates awareness and understanding of your own study.

⬇ How to Handle Problems

Minimise importance if possible

Suggest reasons or offer solutions

⬇ Adds Value to Discussion/Conclusion

Limitations naturally lead to directions for future research.

⬇ Don’t Wait for Perfection

Publishing speed matters — communicate valuable results now, acknowledge limitations openly.

Should Implications & Applications Wait for the Discussion?

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⬇ Implications Begin in Results
Writers often use verbs like *suggest* or *indicate* to guide readers toward the Discussion.

⬇ Transition to Discussion/Conclusion
Mentioning implications signals where the paper is heading.

⬇ Applications Can Appear Early
Connect results back to aims or applications stated in the Introduction/Abstract.

⬇ Echo the Language of Aims
If the aim was "to improve accuracy," use phrases like *more accurate* to show alignment.

⬇ Positive Framing
Applications are often expressed with "**happy words**" that emphasize contribution.

Happy Words

‘happy’ words ☺
able to
accurate
advantage
effective
efficient
excellent
fully
innovation
new
novel
potential
powerful
practical
promising
relevant
robust
simple
straightforward
successful
superior
unique
valuable

'happy' words ☺
accurate
consistent
direct
easy
excellent
important
precise
relevant
robust
satisfactory
simple
suitable
useful

accurate	efficient	reliable	useful
advantageous	excellent	satisfactory	valuable
beneficial	feasible	simple	viable
better	improve	successful	worthwhile
effective	precise	superior	

(i) Positive language (‘happy’ words)

accurate	efficient	reliable	**Verbs/verb phrases:**	prove
advantage	encouraging	robust		provide a first step
appealing	entirely	significant	allow	provide a framework
appropriate	exact	simple	avoid	provide evidence of
attractive	fast	smooth	compare well with	provide insight into
beneficial	favourable	stable	confirm	remove the need for
clear	feasible	straightforward	enable	resolve
comprehensive	flexible	strong	enhance	reveal
convenient	important	successful	ensure	solve
convincing	intuitive	superior	explain	streamline
correct	low-cost	systematic	facilitate	succeed in
cost-effective	new	unambiguous	help to	support
direct	novel	useful	improve	validate
easy	practical	valid	is able to	yield
economical	precise	valuable	offer	
effective	productive	versatile	outperform	
	realistic	viable		
	relevant			

(ii) !-substitutes (‘very happy’ words)

compelling	exciting	powerful	undeniable
crucial	extraordinary	remarkable	unique
dramatic	ideal	superb	unusual
excellent	invaluable	surprising	unprecedented
exceptional	outstanding	striking	unquestionably
	perfect		vital

A Results Model

In Sentences 1 and 2, the writer	repeats/revisits the methods and findings of studies mentioned earlier in the article.
In Sentences 3 and 4, the writer	briefly summarises the method used in this study.
In Sentence 5, the writer	invites the reader to look at a Results graphic.
In Sentences 6 and 7, the writer	compares the results with those in other studies, using subjective, evaluative language.
In Sentence 8, the writer	directs the reader's attention to a specific result, describing it with strongly evaluative language.
In Sentence 9, the writer	selects a specific result to present in more detail, and comments on it.
In Sentence 10, the writer	mentions a possible implication of the results.
In Sentence 11, the writer	mentions a possible limitation, minimising its potential impact on the results.
In Sentence 12, the writer	focuses the reader's attention away from the problem and towards the positive value of the study in terms of the implications and applications of the results.

GENERIC RESULTS MODEL	
1	REVISITING THE LITERATURE/AIM/PREDICTION/HYPOTHESIS/GAP REVISITING/SUMMARISING THE METHOD GENERAL STATEMENT ABOUT THE RESULTS INVITATION TO VIEW GRAPHIC + CONTENT OF GRAPHIC
2	SPECIFIC/KEY RESULTS ± EVALUATIVE LANGUAGE/COMMENTS COMPARISON WITH RESULTS IN OTHER STUDIES/HAPPY WORDS ☺ COMPARISON WITH MODEL/SIMULATION/PREDICTED RESULTS EXPLANATION OF RESULTS VIA KNOWN FACTS/METHOD DETAILS
3	PROBLEMS/ISSUES WITH RESULTS ± REASONS
4	POSSIBLE IMPLICATIONS OF RESULTS/HAPPY WORDS ☺



Using Target Articles to Build a Model

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The Exercise (Step 1)

- Select **at least TWO target articles** from your field → Student Activity 2 (Week 11)
- For each Results section:
  - Identify the **function of each sentence**
- Examples:
  - "Table 1 shows the flow velocity …" → referring to data/figure
  - "This result indicates higher efficiency …" → short interpretation

Using Target Articles to Build a Model

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The Exercise (Step 2)

- Next: analyze the **simplified Results section** provided in the material
- Write a description of the function of each sentence
- Compare your list with the simplified example

Using Target Articles to Build a Model

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The Exercise (Step 3)

- Compare and integrate:
  - **General components** (common across fields)
  - **Field-specific components** (unique to your research area)
- Build a **robust set of model components** for the Results section in your field

“Writing in science is not only about reporting facts,
but about building understanding”

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